

Problem 8.16. Read the definition of *MIN-FORMULA* in Problem 7.46.

Part a. Show that *MIN-FORMULA* $\in$ PSPACE.

Proof. To show that *MIN-FORMULA* $\in$ PSPACE, we give a polynomial space algorithm M .

$M =$ “On input $\langle \phi \rangle$, where ϕ is a Boolean formula:

1. Repeat for every literal a in ϕ :
2. Construct a new Boolean formula ϕ_t by replacing a with *true* in ϕ .
3. Test satisfiability of ϕ_t . Recursively construct every possible assignment and evaluate it. Reuse space for each assignment.
4. Reusing space consumed in step 2, construct a new Boolean formula ϕ_f by replacing a with *false* in ϕ .
5. Test satisfiability of ϕ_f . Reusing space consumed in step 3, repeat step 3 for ϕ_f .
6. If satisfiability of ϕ_t is same as ϕ_f , then *reject*.
7. *Accept*.”

□

Part b. Explain why this argument fails to show that *MIN-FORMULA* $\in$ coNP:

If $\phi \notin \text{MIN-FORMULA}$, then ϕ has a smaller equivalent formula. An NTM can verify that $\phi \in \overline{\text{MIN-FORMULA}}$ by guessing that formula.

Proof. This argument fails to show that *MIN-FORMULA* $\in$ coNP because for *MIN-FORMULA* to be in coNP, an NTM must decide $\overline{\text{MIN-FORMULA}}$ in non-deterministic polynomial time. □